Module 05

1. FTP (File Transport Protocol), Email, and DNS

FTP (File Transfer Protocol)

FTP is a standard network protocol used to transfer files from one host to another over a TCP-based network (like the internet). It operates over the **application layer** of the OSI model and uses ports 20 and 21.

- **Working**: FTP uses a client-server model. The client sends a request to the server for file transfers, and the server responds by providing access to the requested files.
- **Modes**: FTP can operate in two modes:
    - **Active Mode**: The client opens a random port, and the server connects back to the client.
    - **Passive Mode**: The server opens a random port and the client connects to it.
- **Commands**: FTP uses commands like `USER`, `PASS`, `RETR`, and `STOR` to handle the file operations.
- **Security**: By default, FTP is not secure, as it sends data, including passwords, in plain text. Secure variants include **FTPS** (FTP Secure) and **SFTP** (SSH File Transfer Protocol).

Email

Email (Electronic Mail) is one of the oldest and most widely used communication methods on the internet. It enables the exchange of text messages, attachments, and multimedia between users.

- **Protocols**:
    - **SMTP (Simple Mail Transfer Protocol)**: Used for sending emails to mail servers.
    - **IMAP (Internet Message Access Protocol)**: Used for retrieving and managing messages stored on the mail server.
    - **POP3 (Post Office Protocol 3)**: Also used for retrieving email, but it downloads messages and stores them locally, removing them from the server.
- **Working**:
    - SMTP sends emails from the client to the server.
    - IMAP or POP3 retrieves the email from the server to the client.
- **Security**: Common email encryption methods include **TLS/SSL** for secure email transfer.

DNS (Domain Name System)

DNS is like the phonebook of the internet. It translates human-readable domain names (e.g., `www.example.com`) into IP addresses (e.g., `192.168.1.1`) so browsers can load internet resources.

- **Working**: When you type a domain name into your browser, DNS servers are queried to resolve the domain into an IP address. If the information is not cached, DNS queries are made to root servers, then top-level domain servers, followed by authoritative name servers.
- **Types**:
    - **Recursive DNS**: The server will make queries on behalf of the client until it gets an answer.
    - **Iterative DNS**: The client makes multiple requests, each time with more specific information, until the IP is resolved.
- **Security**: DNS can be vulnerable to attacks like DNS spoofing. **DNSSEC (DNS Security Extensions)** can help mitigate these risks by providing data integrity and authentication.

2. Application Layer Architecture

The **Application Layer** is the topmost layer in the OSI model and deals with high-level protocols and services for end-user communication. The architecture of the Application Layer involves interaction between the application software and lower layers of the network stack.

- **Components**:
    1. **Application Software**: Programs such as web browsers, email clients, etc.
    2. **Application Protocols**: Protocols like HTTP, FTP, SMTP, DNS, etc.
    3. **Middleware**: Software that enables communication between applications and the network.
    4. **Client-Server Model**: A widely used model where the client requests services, and the server responds to them.

Neat Diagram:

```
+-------------------------------+
|         Application Layer     |
|     (HTTP, FTP, SMTP, DNS)    |
+-------------------------------+
|      Presentation Layer       |
|     (Data Encoding, Encryption) |
+-------------------------------+
|         Session Layer         |
|     (Session Management)      |
+-------------------------------+
|         Transport Layer       |
|     (TCP/UDP, Error Recovery) |
+-------------------------------+
|         Network Layer         |
|     (IP Routing, Forwarding)  |
+-------------------------------+
|       Data Link Layer         |
|     (MAC, Ethernet, Switching) |
+-------------------------------+
|        Physical Layer         |
|     (Cabling, Signals, etc.)  |
+-------------------------------+
```

- **Functionality**:
    - **Presentation Layer**: Translates data formats between the application and lower layers, ensuring that the data sent to the Application Layer is understandable.
    - **Session Layer**: Manages sessions between devices. It ensures that data is properly synchronized between the application programs.
    - **Transport Layer**: Responsible for end-to-end communication, error correction, and flow control using protocols like TCP and UDP.
    - **Network Layer**: Handles routing and logical addressing (IP), directing packets to the appropriate destination.
    - **Data Link Layer**: Provides node-to-node data transfer, error detection, and correction.
    - **Physical Layer**: Handles the physical transmission of raw data over the medium.

3. Distance Vector Routing Algorithm

The **Distance Vector Routing Algorithm** is a type of routing protocol used in networking, where each router maintains a table (vector) that holds the distance (cost) to every other node in the network. The router periodically shares its routing table with neighbors.

- **Working**:
  1. Each router initially knows only the cost to its directly connected neighbors.
  2. Routers then send their routing tables to their neighbors, which in turn update their own tables based on the information received.
  3. The routers update their tables using the Bellman-Ford equation:
     `D(x) = min(D(x), D(v) + C(v, x))`
     Where `D(x)` is the distance to destination `x`, and `C(v, x)` is the cost to reach node `x` from neighbor `v`.
- **Example**: Consider a simple network with three routers A, B, and C:
  - Initially, Router A knows that the cost to B is 1 and the cost to C is 4.
  - Router B knows that the cost to A is 1 and to C is 2.
  - Router C knows that the cost to A is 4 and to B is 2.

  The routers share their distance tables, and through the updates, Router A learns that the cost to C is 3 (via B), and similarly, other routers update their tables accordingly.
- **Limitations**:
  - **Slow Convergence**: The network takes time to stabilize after a topology change.
  - **Count-to-Infinity Problem**: A router may keep increasing the distance to a destination indefinitely if there's a loop.
  - **Scalability**: Not ideal for large networks because it relies heavily on the periodic exchange of large routing tables.